

Day 3 — Why It Works — The Algebra

Module: $\times 11$ Multiplication — Ekādhikena Pūrveṇa (Primary) · **Time:** 30 min

Goal

Derive $(10a+b)\times 11 = 100a + 10(a+b) + b$ and explain.

Materials

- Notebook, pencils, drawing paper
- (See lesson-plan.md for any day-specific materials)

The flow

Settle + chant (3 min)

Daily chant or breathing exercise.

Story recall (4 min)

Briefly recall yesterday's lesson by asking: “*What’s one thing you remember from yesterday?*”

Today’s idea (10 min)

- *Anurūpyena* (IAST: Anurūpyena; lit. “proportionately” — sūtra used for $\times 12$, $\times 13$ extensions”)
- Why does the magic work?

Show 23×11 with paper strips: - One strip of 23 + ten strips of 23 = 11 strips. - Stack them: $230 + 23 = 253$. - The “230” gives the 2 and the 3 (separated). The “23” gives the +2 in the tens and +3 in the ones. - $2 + 3 = 5$ in the tens place. The 2 and 3 stay on the ends!

Magic? No — just adding! Children try with strips of 14: $14 \times 11 = 140 + 14 = 154$.

Activity (8 min)

A simplified version of *Mental Math Sprint Cards* (see activities/). Draw, sing, or act it out.

Wrap-up (3 min)

Each child shares one word that describes today.

Homework

(See homework.md.)

Teacher notes

- Citations belong in teacher notes only, not in student-facing material at this band.

- Keep new Sanskrit terms to one per lesson, repeated several times.
- If energy is low, swap the activity for free drawing.